

Quiz 9

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Test the following series for convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{2^n}{n!}$ (Using the Ratio Test)

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{2^{n+1}}{(n+1)!}}{\frac{2^n}{n!}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}}{(n+1)!} \cdot \frac{n!}{2^n} \right|$$

$$= \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{2^n}{n!}$ is (absolutely) convergent.

(b) $\sum_{n=1}^{+\infty} \frac{n}{3^n}$ (Using the Root Test)

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left(\frac{n}{3^n} \right)^{\frac{1}{n}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^{\frac{1}{n}}}{3} = \frac{1}{3} < 1$$

Thus $\sum_{n=1}^{+\infty} \frac{n}{3^n}$ is (absolutely) convergent.

Problem 2. (5 points) Test the following alternating series for convergence, absolute convergence or divergence.

(a) $\sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n+1}$

$$a_n = \frac{(-1)^{n+1}}{2n+1} \rightarrow u_n = \frac{1}{2n+1}$$

$$i) u_{n+1} = \frac{1}{2(n+1)+1} < \frac{1}{2n+1} = u_n$$

$$ii) \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{1}{2n+1} = 0$$

$\Rightarrow \sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n+1}$ is convergent

by the Alternating series Test

$$\sum_{n=1}^{+\infty} |a_n| = \sum_{n=1}^{+\infty} \frac{1}{2n+1} \text{ is divergent}$$

So $\sum_{n=1}^{+\infty} \frac{(-1)^{n+1}}{2n+1}$ is only conditionally convergent.

(b) $\sum_{n=1}^{+\infty} (-1)^{n+1} \frac{3n}{2n+1}$

$$a_n = (-1)^{n+1} \frac{3n}{2n+1} \rightarrow u_n = \frac{3n}{2n+1}$$

$$\lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{3n}{2n+1} = \lim_{n \rightarrow \infty} \frac{3}{2+\frac{1}{n}} = \frac{3}{2}$$

Thus

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (-1)^n u_n \text{ doesn't exist}$$

$\Rightarrow \sum_{n=1}^{+\infty} (-1)^{n+1} \frac{3n}{2n+1}$ is divergent.

by the Divergence Test.